

Feature	for loop	while loop
Purpose	Used when the number of iterations is known or can be determined in advance.	Used when the number of iterations is unknown and depends on a condition .
Condition Type	Works with a sequence (like list, range, string, tuple, etc.).	Works with a logical condition (True/False).
Syntax	for variable in sequence: # code block	while condition: # code block
Automatic Counter	Has a built-in counter via the sequence (like for i in range(5)).	Needs a manual counter that you must update (i += 1).
Best Used When	You know how many times to repeat the loop.	You want to repeat until a certain condition becomes false.
Example	python for i in range(5): print(i)	python i = 0 while i < 5: print(i) i += 1
Risk of Infinite Loop	Less likely, as it automatically stops after iterating the sequence.	More likely if the condition never becomes False.
Else Clause	Can have an else part executed when the loop completes normally.	Also supports an else part executed when the condition becomes false.